a **practical 433 MHz IQ modulator (1 MHz BW)** using **dual-gate MOSFET mixers** (I/Q) + **quadrature LO** + **433 MHz band-pass** + a **small linear RF gain block** to reach **+10 dBm** into 50 Ω.

> Reality check: getting clean **+10 dBm** *linear* output at 433 MHz is much easier/reliable with an **MMIC gain block** (or a proven RF driver IC) than with a fully-discrete MOSFET PA. The "IQ modulator" part is MOSFET-based as you asked; the output stage is a clean linear amplifier.

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433 MHz IQ Modulator +10 dBm (1 MHz BW) — Schematic

1) Quadrature LO generation (0° / 90°)

Use a stable 433.92 MHz LO source (PLL/VCO/TCXO+multiplier) → buffer → 90° hybrid.

```
                +5V
                 |
            [LO BUFFER]
     LO_IN o--CLO1--+---------------------> J1 LO_BUF_OUT (~+7 to +10 dBm)
            100p    |
                  GND

J1 LO_BUF_OUT --> 90° HYBRID COUPLER @ 433 MHz (3 dB quadrature)
                  ┌──────────────────────────────┐
                  │   IN    -> LO_IN             │
                  │   OUT0  -> LO_I   (0°)       │----> LO_I
                  │   OUT90 -> LO_Q   (90°)      │----> LO_Q
                  │   ISO   -> 50Ω to GND        │
                  └──────────────────────────────┘
```

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2) I and Q mixers (dual-gate MOSFET, e.g., BF998/BF999)

Each mixer is identical; only LO phase and baseband input differ.

I-Mixer (U1) and Q-Mixer (U2)

- **Gate2** = LO injection
- **Gate1** = baseband I or Q (biased for linearity)
- **Drain** = RF current output into tuned/loaded node

```
                        +5V
                         |
```

```
                          RFC_D1   220 nH
                            |
                            +--------o DRAIN U1 (BF998)     I-path RF current
                            |
                           CDD1  100n || 1n || 10p  (decouple close)
                            |
                           GND

U1 BF998 (Dual-gate MOSFET mixer)
---------------------------------
SOURCE U1 --- RS1 33Ω --- GND
              |
             CS1 100pF to GND (RF bypass)

GATE2 (LO_I) injection + bias:
LO_I ---- Cg2I 10pF ---- G2(U1)
                 |
               Rg2I1 10k to +2.5V
               Rg2I2 10k to GND
               Cb2I  1nF to GND (at G2 bias node)

GATE1 (I baseband) injection + bias:
I_IN --- RinI  1k ---+--- G1(U1)
                     |
                   Rg1I1 47k to +1.2V (BIAS_I)
                   Rg1I2 47k to GND
                   Cb1I  10nF to GND (at G1 bias node)
```

Duplicate for Q:

- U2 BF998
- Gate2: LO_Q via 10 pF
- Gate1: Q_IN via 1 k, with same bias network

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3) Drain combining + 433 MHz band-pass selection (1 MHz BW)

Combine drain currents into a common **RF_SUM** node, then a **433 MHz band-pass** to suppress LO leakage / images and set channel.

A simple, buildable approach:

- Give each drain a **small series resistor** for combining stability
- Use a **single parallel resonator** at RF_SUM (tuned to 433.92 MHz)
- Then a **2–3 pole band-pass** (recommended) or at least a low-pass

```
U1 DRAIN --- Rd1 22Ω ---+
                        +---- RF_SUM -----> 433 MHz BPF -----> RF_DRV_IN (50Ω)
U2 DRAIN --- Rd2 22Ω ---+

At RF_SUM (tuning / select):
                  LSUM  68 nH
RF_SUM ----+-----^^^^-----+---- (to BPF)
           |              |
          Ctrim          Csh
      (2–6 pF trimmer)  3.3 pF
           |              |
          GND            GND
```

Band-pass filter (recommended): Use a small 433 MHz SAW filter (best) or a 3-pole LC BPF.

- For **1 MHz modulation bandwidth**, choose a filter bandwidth comfortably larger (e.g., **5–10 MHz**) to avoid distorting the modulation.

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4) Linear RF gain block to reach +10 dBm

After the BPF, add a **broadband 50 Ω gain block** (MMIC) with ~15–20 dB gain and P1dB > +12 dBm.

```
RF_DRV_IN (50Ω) --> C1 100p --> [MMIC GAIN BLOCK] --> C2 100p --> 433 LPF/BPF --> OUT 50Ω
                                  | Vdd=+5V
                                  +-- RFC 220 nH -- +5V
                                  +-- decoupling: 100n || 1n || 10p to GND
```

Then a final **low-pass** or **band-pass** at the output for spectral cleanliness.

Baseband drive (I/Q, 1 MHz BW)

You'll want low-impedance baseband sources (DAC or op-amp drivers) into Gate1 networks.

Typical baseband driver per channel:

```
DAC_I/OPAMP_I --> 50–200Ω series --> I_IN (to RinI 1k)   (limit ringing, protect)
```

```
DAC_Q/OPAMP_Q --> 50–200Ω series --> Q_IN
```

Keep baseband traces short; decouple the bias nodes well.

Notes that matter for your case (+10 dBm, 1 MHz BW)

- **LO level** into each MOSFET Gate2 usually needs a few dBm (often **0 to +7 dBm** effective at the gate after coupling). Too low → poor conversion; too high → distortion/LO feedthrough.
- **Bias tuning is critical**: Gate1 bias (~1.0–1.5 V) and Gate2 bias (~2–3 V) set linearity vs conversion gain.
- A **SAW filter** at 433 MHz is the easiest way to clean LO leakage and images.
- Your **layout** is everything at 433 MHz: symmetric LO routing to I/Q mixers, short drain combine, solid ground stitching, and tight decoupling.